Chapter 6: Samples and Observational Studies

The Practice of Statistics in the Life Sciences
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Part III: Data Generation

Why This Chapter Matters

Statistical methods are useful only when the data have been collected in a way that matches the scientific question.

Before a mean, proportion, confidence interval or test is interpreted, the study design should make clear

- who or what is being studied
- what variables are being measured
- how the observations are obtained

The quality of a statistical conclusion depends on the quality of the data-generation process.

Gathering Data and Study Design

- Many scientific questions concern populations that are too large to study completely.
  1. A health researcher may want to estimate a proportion for adults in a country.
  2. A wildlife biologist may want to learn about fish in a river system.
  3. A laboratory may want to understand a biological process from a manageable set of specimens.
- In each setting, the main issue is not simply how much data are collected. The main issue is whether the data were collected through a design that supports the intended conclusion.
- A numerical method cannot repair a study design that systematically excludes important groups, measures the wrong variable or mixes the effect of an explanatory variable with other factors.
- Statistical reasoning therefore begins with a clear description of how the data were generated.

Observational Studies versus Experiments

Suppose researchers want to know whether a fertilizer increases seedling growth.

Observational Study: Observe Existing Conditions

Researchers compare seedlings that growers already fertilize with seedlings they do not fertilize. The researchers record conditions and growth without assigning fertilizer use.

Experiment: Assign Treatments

Researchers assign seedlings to receive fertilizer or water alone then compare their growth over four weeks.

In the observational study, the groups might also differ in soil, sunlight, or watering. These differences could help explain any observed difference in growth.

Random Sampling versus Random Assignment

Suppose a greenhouse contains 1000 seedlings.

Random Sampling: Which Seedlings Enter the Study?

Researchers randomly select 60 seedlings from the greenhouse.
Random sampling helps the sample represent the greenhouse population, supporting generalization to that population.

Random Assignment: Which Treatment Does Each Receive?

Researchers randomly assign the selected seedlings: 30 receive fertilizer and 30 receive water alone.
Random assignment makes the treatment groups comparable on average, supporting a causal comparison.

A study can use one, both, or neither. Random assignment does not guarantee that the study participants represent a wider population.

When Are Observational Studies Needed?

A well-designed randomized experiment strengthens causal interpretation but researchers cannot always assign the conditions they want to study.

- **An experiment would be unethical.** Researchers should not assign people to smoke cigarettes to investigate long-term health outcomes.
- **The condition cannot be assigned.** Researchers studying age and memory must observe people of different ages.
- **The research concerns existing conditions.** Ecologists may compare wildlife populations in habitats that have already experienced different levels of disturbance.

How Can Researchers Strengthen the Evidence?

They can measure other relevant variables, compare similar groups and examine alternative explanations for the association.

Lurking Variables: What Else Might Explain Growth?

Suppose an observational study finds that fertilized seedlings grow more than unfertilized seedlings.

- The explanatory variable is **fertilizer use**.
- The response variable is **seedling growth**.

However, growers who use fertilizer might also place their seedlings in sunnier locations. The researchers did not record sunlight exposure.

Lurking Variable

A lurking variable is a variable omitted from the analysis that may help explain the observed relationship. Here, **sunlight exposure** is a possible lurking variable.

Greater growth could reflect fertilizer, sunlight or both.

Confounding: Which Factor Explains the Difference?

Now suppose nearly all fertilized seedlings receive full sunlight while nearly all unfertilized seedlings grow in shade.
The groups differ in **two ways**: fertilizer use and sunlight. The data provide little opportunity to compare fertilizer treatments under the same sunlight conditions.

Confounding

Confounding occurs when the effects of different explanatory variables on the response cannot be clearly separated. Here, the effects of fertilizer and sunlight are confounded.

How the Ideas Differ

"Lurking" describes a variable's omission from the analysis. "Confounding" describes the difficulty of separating effects. Even a recorded variable can be involved in confounding.

Reducing Confounding in Observational Studies

To investigate fertilizer use, researchers could make comparisons among seedlings grown under similar conditions.

- **Restriction:** include only seedlings grown in full sunlight.
- **Matching:** compare fertilized and unfertilized seedlings with similar sunlight exposure and soil conditions.
- **Stratification:** compare fertilizer groups separately within full-sun, partial-shade, and shade categories.
- **Statistical adjustment:** include measured factors, such as sunlight and watering, in a statistical model.

A Limitation

These methods can reduce confounding when suitable comparisons are available. Unmeasured or poorly measured differences can still affect the conclusions.

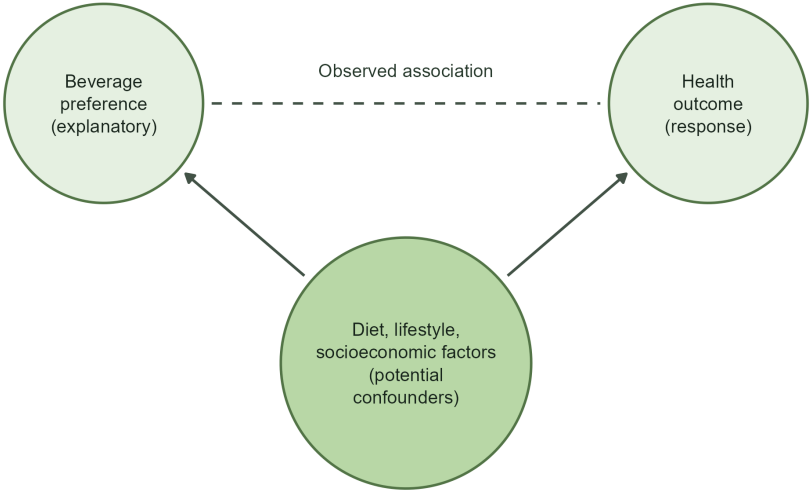
Example 6.2: Wine, Beer or Spirits?

Question

Some observational studies have reported that health outcomes differ among moderate drinkers who prefer wine, beer or spirits. Does an observed difference in health establish that beverage type itself causes the difference?

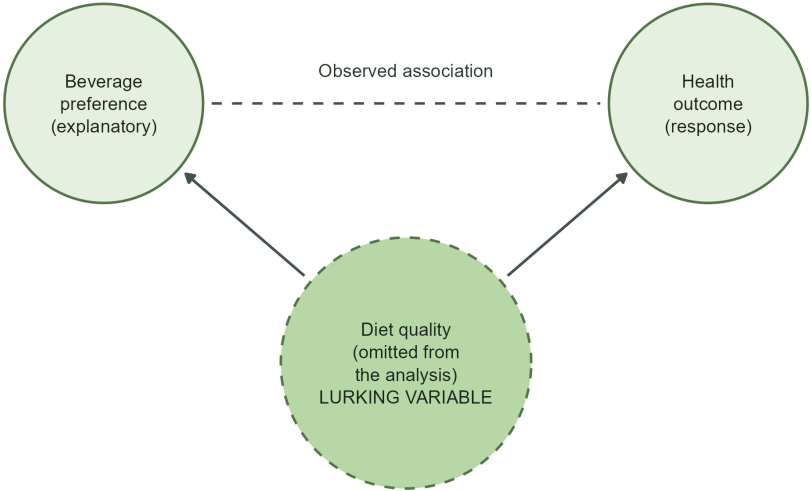
- Beverage preference can be related to many other characteristics.
- Groups that prefer different beverages may also differ in diet, smoking behaviour, exercise, income, education and access to health care.
- If these characteristics also relate to the health outcome, the effect of beverage preference cannot be separated cleanly from the effects of the other variables.

Example 6.2: Wine, Beer or Spirits?



Solid arrows show possible influences, not established causal effects.

Example 6.2: Wine, Beer or Spirits?



Possible influences are shown. The dashed border marks a variable omitted from the analysis.

Example 6.2: Wine, Beer or Spirits?

- The observational association does not by itself establish a causal effect of beverage type.
- Potential confounders provide alternative explanations for the observed difference.
- A scientifically appropriate conclusion is that beverage preference and health are associated in the observed data after considering the study design. A stronger causal claim would require stronger evidence and careful control of competing explanations.

Sampling

A university health centre wants to learn about the sleep habits of first-year students. Asking every first-year student would be a **census**. Instead, the centre could collect information from **sample**: a subset of those students.

Why Sample?

A census can be expensive, slow or difficult to complete. A carefully selected sample can provide useful information with fewer resources.

The main challenge is deciding **how to select students** so that the results help us understand the larger group.

Example: Defining the Population and Variable

The health centre wants to estimate the proportion of currently enrolled first-year students who average at least 7 hours of nighttime sleep.
For this study, each student reports their average hours of night time sleep over the previous 7 nights.

Questions

1. What is the population of interest?
2. What variable determines whether a student belongs to the group being counted?

Example: Defining the Population and Variable

Population

All currently enrolled first-year students at that university. This is the group the health centre wants to describe.

Variable

Whether a student's reported average night time sleep over the previous 7 nights is at least 7 hours:

$$\text{Yes: average} \geq 7 \qquad \text{No: average} < 7$$

This is a **categorical variable**.

The reported average number of hours is a quantitative variable. We classify it into yes or no categories to answer the study's question.

From the Population to the Sample

The health centre needs a practical way to identify and select students.

Sampling Frame

The list or mechanism used to identify units that can be selected. For example, it could be the university's current first year enrolment list.

Sampling Design

The planned method for selecting units. For example, randomly select 200 distinct student IDs from that list and invite those students to complete the survey.

The **sample** provides the data. If some selected students do not respond, the health centre must consider whether respondents differ from nonrespondents.

Parameter versus Statistic

Parameter: Describes the Population

Let p be the proportion of **all first year students** at this university who meet the sleep criterion. This population proportion is unknown.

Statistic: Calculated from the Sample

Suppose all 200 selected students respond, and 124 meet the sleep criterion. The sample proportion is

$$\hat{p} = \frac{124}{200} = 0.62$$

Another sample could produce a different value of $\hat{p}$.

Statistical inference uses the observed 62% to estimate p and describe the uncertainty in that estimate.

Why the Selection Method Matters

Suppose the health centre surveys only students studying in the library at 10 p.m.

Who Might Be Missed?

Students who study elsewhere, work evening shifts, commute or go to bed earlier may be absent. Their sleep habits could differ from those of students in the library.

Selecting students randomly from a complete enrolment list gives students across these groups a chance of selection.

Sample Size Is Not Enough

Surveying more students in the same library does not remove the problem of systematically missing other students.

Planning the Sleep Survey

Before collecting data, the health centre specifies:

1. **Target population:**currently enrolled first year students at this university.
2. **Measurement:** reported average nighttime sleep over the previous 7 nights, classified using the 7-hour threshold.
3. **Sampling frame:** an updated first year enrolment list.
4. **Selection method:** randomly select 200 distinct student IDs.
5. **Missing-data plan:** arrange reminders, record response rates and specify how incomplete answers will be handled.

Generalizing from the Sample

Suppose 62% of the surveyed students meet the sleep criterion. What can the health centre conclude?

- The result estimates the proportion among **first year students at this university**.
- The design does not automatically support conclusions about students in other years or at other universities.
- Even with random selection, the estimate can vary from sample to sample.
- An incomplete enrolment list, nonresponse, or inaccurate sleep reports can introduce additional errors.

The Main Principle

A useful generalization depends on who was eligible, how students were selected and how the data were collected.

Sampling Variability versus Bias

Suppose a university repeatedly selects samples of first year students to estimate the proportion meeting a sleep criterion.

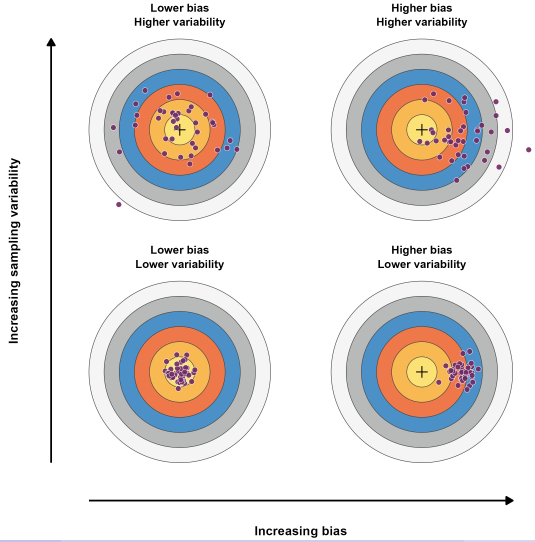
Sampling Variability: How Much Do Estimates Differ?

Different random samples usually produce different sample proportions. This natural variation from sample to sample is **sampling variability**.

Bias: Where Are the Estimates Centred?

A method is biased if its estimates systematically tend to be above or below the true population value.
For example, surveying only students in the library late at night may systematically misrepresent the university's first year students.

Archery Analogy: Bias and Sampling Variability



Reading the Four Targets

The horizontal axis compares **bias**. The vertical axis compares **sampling variability**.

- **Upper left:** estimates vary widely but are centred near the truth.
- **Upper right:** estimates vary widely and are centred away from the truth.
- **Lower left:** estimates are tightly grouped near the truth.
- **Lower right:** estimates are tightly grouped but systematically miss the truth.

Two Different Questions

How spread out are the estimates? Where is the distribution of estimates centred?

What Does Increasing Sample Size Improve?

Under the same sampling design, larger samples generally reduce sampling variability.

In the Archery Analogy

Increasing sample size can tighten the grouping of estimates. It does not necessarily move that grouping toward the bullseye.

For example, surveying more late night library users may give a more precise estimate for that group while still misrepresenting all first year students.

Precision Does Not Guarantee Accuracy

A tightly grouped set of estimates can still be systematically wrong. Good sampling requires attention to both variability and bias.

Poor Sampling Designs: Who Gets Included?

The health centre wants to estimate the proportion of first-year students who average at least 7 hours of night time sleep. Consider two ways it might collect responses:

Plan A: Ask Whoever Is Nearby

A researcher surveys students leaving a first year lecture. This is a **convenience sample**.

Plan B: Invite Anyone Interested

The centre posts an open survey link: "Are you getting enough sleep? Tell us about it!" This is a **voluntary response sample**.

In both cases, the participating students may differ systematically from the population the centre wants to describe.

Convenience Sampling: Easy to Reach

Definition

A convenience sample selects individuals or units because they are readily available.

Surveying students leaving one lecture is convenient but the sample includes only students who attend that class.

- Students in other courses have no opportunity to be included.
- Students absent from that lecture are missed.
- Attendance and course schedules may be related to sleep habits.

Why This Matters

Even a large number of responses from that lecture may misrepresent first year students across the university.

Voluntary Response: Who Chooses to Participate?

Definition

A voluntary response sample forms when an open invitation allows individuals to select themselves into the sample.

For the online sleep survey, students who are concerned about their sleep may be especially motivated to respond.

- Participation may depend on the very characteristic being studied.
- Posting the link widely does not ensure that respondents represent all first-year students.

An Important Distinction

An open survey is voluntary response sampling. Inviting randomly selected students is a different design.

Probability Samples: Selection by Chance

Veterinary researchers want to estimate the average weight of all dairy cows in a region. They cannot weigh every cow so they need a sample.

Probability Sample

A probability sample uses a random selection procedure. Every population unit has a **known nonzero probability** of being included.

For example, researchers could randomly select cow IDs from a complete regional registry.

- Researchers follow a defined selection procedure.
- Selection probabilities need not be equal.
- The design allows sampling uncertainty to be assessed.

Simple Random Sampling: Choosing Individual Cows

Suppose the registry lists 1000 dairy cows. Researchers want to select 50 distinct cows.

Simple Random Sample

In a simple random sample of size n, every possible group of n population units has the same chance of selection.

The researchers could number the cows from 1 to 1000 and use software to select 50 distinct numbers at random.

$$P(\text{a particular cow is selected}) = \frac{50}{1000} = 0.05$$

Every cow has the same inclusion probability and every possible group of 50 cows is equally likely. Selecting nearby cows until 50 are reached would not satisfy this definition.

Other Probability Designs: Sampling Across Farms

Selecting individual cows across an entire region may require considerable travel. Other designs organize selection differently.

- **Stratified sampling:** divide the registry into breed groups and select a random sample of cows from *every* breed group.
- **Cluster sampling:**randomly select farms and weigh *every* dairy cow on the selected farms.
- **Multistage sampling:** randomly select farms, then randomly select cows *within* those farms.

The Main Difference

Stratification samples within every group. Cluster sampling selects some groups. Multistage sampling adds further selection within chosen groups.

Random Sampling versus Random Assignment

Suppose the researchers also want to compare two feed supplements.

Random Sampling: Which Cows Enter the Study?

Randomly select cows from the regional registry.
This supports generalization from the sampled cows to the population covered by that registry.

Random Assignment: Which Supplement Does Each Receive?

Randomly assign the participating cows to supplement A or supplement B, then compare their weight gain.
This makes the treatment groups comparable on average and supports a causal comparison.

A study can use both procedures. Random assignment alone does not make the participating cows representative of the region.

Break Time



Sample Surveys

A sample survey is a study that gathers information from a sample in order to describe a population.

Even a large survey can be misleading if some groups are missed, many selected individuals do not respond or the questions are poorly designed.

Undercoverage

Undercoverage occurs when some groups in the population are left out of the sampling frame or are given little chance of selection.

A survey that samples only landline telephone numbers, for example, can underrepresent people who rely mainly on cell phones.

Example 6.8: The Cell Phone Problem

Question

A health survey uses only landline telephone numbers to sample adults. Why can this design produce undercoverage bias?

Example 6.8: Solution

Adults who rely mainly on cell phones may be missed or underrepresented.

If cell-phone-only adults differ from landline users on the characteristic being measured, the sample statistic can be systematically different from the population value.

The problem is not simply that the sample is smaller. The problem is that the sampling frame excludes part of the target population.

Nonresponse Bias

Nonresponse bias occurs when selected individuals do not participate and the nonrespondents differ in important ways from those who do respond.

Weighting or follow-up efforts may reduce the problem, but they do not automatically remove it.

Response Bias

Response bias occurs when the answers given are systematically inaccurate.

Sensitive topics, interviewer effects, question wording and social pressure can all affect how people answer.

Wording and Order of Questions

The wording and order of survey questions can influence the responses.

Leading wording can push respondents toward a particular answer. Earlier questions can also shape how later questions are interpreted.

Example 6.10: Question Order and Reported Happiness

Question

A survey first asks people about recent dating experiences and then asks how happy they are overall. Why might the answer to the second question be affected by the first?

Example 6.10: Solution

The first question can bring a specific aspect of life to mind and make it unusually prominent when the respondent answers the second question.

The overall happiness response may therefore reflect the temporary influence of the earlier question rather than the person's broader long-run judgment.

A Survey Quality Checklist

When evaluating a survey, useful questions include

- Who is the target population
- What is the sampling frame
- How were units selected
- What was the response rate
- Could wording or ordering influence the answers

Cohort and Case-Control Studies

Case-control and cohort studies are observational designs used to study relationships between exposures and outcomes.

They are especially important when an experiment would be unethical, impractical or too slow.

Case-Control Studies

A case-control study selects individuals according to their outcome status.

The cases have the outcome of interest. The controls do not and should ideally come from the same source population.

This design is especially useful for rare outcomes because researchers do not need to wait for enough cases to appear in a very large cohort.

Case-Control Studies: What They Can and Cannot Do

Case-control studies can efficiently examine whether previous exposures were more common among cases than among controls.

They cannot directly estimate population prevalence or incidence from the researcher-chosen case-control ratio.

Real-World Case Study: Reye's Syndrome and Aspirin Exposure

Question

Why is a case-control design efficient for a rare outcome such as Reye's syndrome? What population quantity cannot be estimated directly from the researcher-chosen case-control ratio?

Real-World Case Study: Explanation

The design begins by locating cases, so it does not require a huge random sample while waiting for a rare outcome to occur.

Controls provide a comparison group for examining whether previous exposures were more common among cases.

Real-World Case Study: Solution

The design is efficient because it concentrates data collection on individuals who are informative about a rare outcome.

It cannot directly estimate population prevalence or incidence from the case-control sampling ratio.

It can, however, estimate measures of association between exposure and outcome when the cases and controls are selected appropriately.

Cohort Studies

A cohort study identifies a group of individuals according to exposures or characteristics, then compares the occurrence of outcomes across those groups.

A prospective cohort collects exposure information before the outcomes occur and follows participants forward through time.

Strengths and Limitations of Cohort Studies

Cohort studies can estimate incidence within the observed cohort, compare outcome rates across exposure groups and study several outcomes from the same exposures.

Their main practical disadvantages include time, cost and loss to follow-up.

Their main statistical limitation is that exposure groups are not created by random assignment, so confounding can remain.

Real-World Case Study: The British Doctors Study

Question

What information can a cohort design provide in a study of smoking and later health outcomes? What limitations remain?

Real-World Case Study: Explanation

A prospective cohort records exposure information before many outcomes occur, which clarifies the time order between exposure and outcome.

Because the cohort is followed over time, researchers can estimate rates of new outcomes and compare those rates across exposure groups.

Real-World Case Study: Solution

A cohort study can estimate incidence within the observed cohort, compare outcome rates across exposure groups and study several outcomes from the same exposures.

Confounding can still remain because the exposure groups were not formed by random assignment.

Attrition can also create selection bias if loss to follow-up is related to both exposure and outcome.

Comparing the Observational Designs

Feature	Sample survey	Case-control study	Cohort study
Typical starting point	A sample from a population	Outcome status	A defined cohort and exposure information
Main direction	Usually a population snapshot	Often looks backward for exposure	Often follows outcomes forward
Useful for rare outcomes	Usually inefficient	Especially useful	Can require a very large cohort
Can estimate prevalence directly	Yes, with appropriate sampling and measurement	Not from the chosen case-control ratio	Sometimes, depending on the cohort and timing
Can estimate incidence directly	Usually not from one cross-sectional survey	No	Yes, within a properly followed cohort
Main concerns	Coverage, nonresponse and response bias	Selection, recall and confounding	Confounding, attrition and cost

From Observational Studies to Experiments

Observational studies can provide strong and scientifically important evidence, especially when experiments are impossible or unethical.

Their interpretation must still consider alternative explanations such as confounding and selection bias.

A randomized experiment addresses a different problem by assigning treatments through chance, which strengthens causal interpretation.

Chapter Summary: Core Ideas

Data collection determines what a statistical analysis can support.

An observational study measures naturally occurring variables without deliberately assigning the main explanatory condition. An experiment deliberately assigns treatments.

Probability sampling connects a sample to a population. Increasing sample size can reduce sampling variability, but it does not automatically remove bias.

Case-control and cohort studies are important observational designs with different strengths and limitations.

A Consistent Study-Design Workflow

A first-year analysis of study design becomes easier when the same questions are asked in a consistent order.

1. What is the scientific question
2. What is the target population
3. What are the explanatory and response variables
4. Is the design observational or experimental
5. If sampling is used, what frame and selection method were used
6. Could bias or confounding affect the conclusion
7. Does the final conclusion match what the design can support